

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)			profile with start and end at same energies and activation energy of 58 kJ mol^{-1} allow tolerance ± 1 small square each way for peak maximum		1		1		
	(b)	(i)		$k = A e^{-E_a/RT}$ accept $\ln k = \ln A - \frac{E_a}{RT}$	1			1	1	
		(ii)		$e^{-E_a/RT} = 1.19 \times 10^{-4}$ (1) ecf possible from part (i) $E_a = 22.5 \text{ kJ mol}^{-1}$ (1) activation energy is lower so this catalyst is more effective (1) ecf from activation energy calculated			3	3	2	
	(c)			$\Delta_f H^\ominus = 2 \times \Delta_f H^\ominus(\text{H}_2\text{O}) - 2 \times \Delta_f H^\ominus(\text{H}_2\text{O}_2)$ (1) $2\Delta_f H^\ominus(\text{H}_2\text{O}_2) = (2 \times -286) - (-98) = -474$ $\Delta_f H^\ominus(\text{H}_2\text{O}_2) = -237$ (1)		1 1		2	1	

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	(d)			positive because the reaction forms a gas (and gases have a higher entropy than liquids)		1				
	(e)	(i)	I	A HCl/H ⁺ (aq) (1 mol dm ⁻³) B platinum/Pt (electrode) both required	1			1		1
			II	electrons flowing from standard hydrogen electrode to dichromate half cell		1		1		1
			III	salt bridge completes circuit without allowing solutions to mix	1			1		1
			IV	this makes the reading less positive/smaller (must give reason to gain this mark) (1) increase in concentration of chromium(III) ions will shift equilibrium to the left (1)			1 1	 2		

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		(ii)	I	$2\text{Cr}^{3+} + 3\text{H}_2\text{O}_2 + \text{H}_2\text{O} \rightarrow \text{Cr}_2\text{O}_7^{2-} + 8\text{H}^+$		1		1		
			II	reaction is feasible (must give reason to gain this mark) (1) as standard electrode potential for hydrogen peroxide is more positive than that for chromium/dichromate / EMF calculated as +0.44V and positive value means reaction is feasible (1)		1	1	2		
			III	electrochemical methods are better (must give reason for this mark) (1) because the reaction is in solution but standard enthalpies of formation use standard states			2	2		
				Question 11 total	3	7	8	18	4	3